Concept of LOSS Function

$$Let\ \mathbf{Y} = \begin{pmatrix} y_1 \\ y_2 \\ \cdots \\ y_n \end{pmatrix}_{n \times 1}$$ **Original Labels** from Training Samples

$$Let\ \hat{\mathbf{Y}} = \begin{pmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \cdots \\ \hat{y}_n \end{pmatrix}_{n \times 1}$$ **Predicted Labels** from Model $\hat{Y}=X\underline{W}$

Remember

$$\|a\|_2 = a^T a$$

Objective is $\hat{Y}=Y$

If $\hat{Y} \neq Y \Rightarrow$ There is an **Error (Loss)**

$\underline{\hat{Y}} = X\underline{W}$ **Predicted**

Where: X_{N×(d+1)} is a matrix ⇒ Given

Y_{N×1} is labels vector ⇒ Given

W is Model Parameter ⇒ Required

Define **Loss** function as **squared distance** between $\hat{Y}$ and Y

$$Loss = \|Y - \hat{\mathbf{Y}}\|_2 = (Y - \hat{\mathbf{Y}})^T (Y - \hat{\mathbf{Y}})$$